

Orthemology notation gallery

Status: DRAFT — not peer reviewed. Rendered by the R7B mathematical- typesetting pipeline (Decision 0023). Every normative symbol in docs/notation-registry.yaml appears below as real mathematics: GitHub renders the `...` inline math and fenced `math` blocks through MathJax, and `scripts/build_pdfs.py` renders artifacts/notation-gallery.pdf through the pinned Typst compiler via `scripts/latex_to_typst_math.py`. This is a **typography layer, not a notation redesign**: each row’s meaning is the registry role, unchanged (Decision 0005). `scripts/validate_math_source.py` asserts a bijection between non-retired symbols and this gallery and that `scripts/validate_pdf_math.py` finds no missing-glyph (notdef/tofu) in the rendered PDF — the defect reproduced in docs/project-closure/r7b/R7B-PDF-MATH-BASELINE.md (the core μ -vector and $\mathbf{C}$ -vector rendering as notdef under the old `#raw` path). Below they compose correctly as $\vec{\mu}$ and $\vec{C}$.

1. Normative symbols rendered as mathematics

Registry key: A
Rendered: A
Role: declared analysis (identifier + version)

Registry key: alpha, beta, ...
Rendered: α, β
Role: actor indices

Registry key: T = task(A)
Rendered: $T = \text{task}(A)$
Role: task component of the analysis

Registry key: M.univ, O.univ
Rendered: $\mathcal{M}, \mathcal{O}$
Role: universal occurrence and ortheme universes

Registry key: M.A, O.A
Rendered: $\mathcal{M}_A \subseteq \mathcal{M}, \mathcal{O}_A \subseteq \mathcal{O}$
Role: analysis-active domain and repertoire

Registry key: Inst.A
Rendered: $\text{Inst}_A \subseteq \mathcal{M}_A \times \mathcal{O}_A$
Role: analysis-relative instantiation (the primitive)

Registry key: O*(m; A)
Rendered: $O^*(m; A)$
Role: true (actual) profile of m under A

Registry key: Pi.A
Rendered: Π_A
Role: complete-profile space under A

Registry key: Pi.A.partial
Rendered: Π_A^∂
Role: partial-profile space under A

Registry key: C.hat
Rendered: $\hat{C}_{A,\alpha,t}(m)$
Role: candidate set of complete profiles

Registry key: p.hat
Rendered: $\hat{p}_{A,\alpha,t}(m) \in \Pi_A^\partial$

Role: inferred (partial) profile — belief, never ground truth

Registry key: Omega
Rendered: $\Omega_{A,\alpha,t}$
Role: observation map

Registry key: mu
Rendered: $\mu = \langle g, S_\mu, \text{select}_\mu, \text{prov}, \text{ver} \rangle$
Role: metaortheme type

Registry key: pi.mu
Rendered: π_μ
Role: paired meta-policy

Registry key: mu.bar
Rendered: $\bar{\mu}_{e,j}$
Role: the j-th metaorthemma in episode e

Registry key: MetaInst
Rendered: $\text{MetaInst}(\bar{\mu}_{e,j}, \mu)$
Role: token typing of a metaorthemma under a metaortheme

Registry key: Gamma.E
Rendered: $\Gamma_E = (E, \rightsquigarrow)$
Role: episode DAG

Registry key: e.Gamma
Rendered: $e_\Gamma = \text{comp}(\Gamma_E)$
Role: composed boundary-level episode

Registry key: GoalSchema
Rendered: $\text{GoalSchema}(\alpha)$
Role: parametric goal schema

Registry key: G.target
Rendered: $\mathcal{G}_{\alpha,A_\alpha}$
Role: actor-alpha’s grounded target profile set

Registry key: Rep.A
Rendered: Rep_A
Role: representation family under A

Registry key: chi
Rendered: $\chi \in \text{Rep}_A$
Role: one representation

Registry key: L_A_star
Rendered: $L_A^*(\chi)$
Role: best attainable loss/risk under representation χ
Registry key: select_mu
Rendered: select_μ
Role: metaortheme state-selecting evidence procedure
Registry key: estatus_e
Rendered: estatus_e
Role: per-claim evidence-status map of episode e
Registry key: eps_A
Rendered: ϵ_A
Role: accepted tolerance of the analysis
Registry key: theta_stop
Rendered: θ_{stop}
Role: ANDON stop/escalation risk threshold
Registry key: K_A
Rendered: $\mathcal{K}_A$
Role: cause repertoire
Registry key: R_A
Rendered: $\mathcal{R}_A$
Role: route repertoire
Registry key: W_A
Rendered: $\mathcal{W}_A$
Role: warrant-state repertoire
Registry key: Q_e
Rendered: $\mathcal{Q}_e$
Role: claim ledger of episode e
Registry key: delta_e
Rendered: δ_e
Role: residual-disposition map of episode e
Registry key: ReqPath
Rendered: $\text{ReqPath}(e)$
Role: governance-derived required pathway verdicts

2. Structural constructs (typesetting showcase)

The pipeline handles the full range of publication constructs the corpus uses. These are the constructs named in the Decision 0023 design spike.

2.1 Episode signature (display)

The formal-core episode signature — the exact site of the reproduced notdef defect — rendered as real mathematics (the vectors $\vec{\mu}$, $\vec{C}$ compose):

$$\begin{aligned}
e = \langle & \text{id}; m, \kappa, v; x, H; \alpha, w, A, T, \\
& t; \vec{\mu}, \text{MetaTok}, \pi; \vec{C}, \hat{p}; r; \\
& \text{estatus}; \mathcal{Q}; \delta; \\
& \text{hand_in}, \text{hand_out}; a, \\
& \text{Succ} \rangle
\end{aligned}$$

2.2 Multi-line aligned equations

$$\begin{aligned}
\text{TokenAdequate}(\bar{\mu}, e) &\Leftrightarrow \text{MetaInst}(\bar{\mu}, \mu) \\
&\quad \wedge \text{Compatible}(\bar{\mu}, A(e)) \\
\text{V3c}(e) &\Leftrightarrow \\
&\quad \forall \bar{\mu} \in \text{MetaTok}(e) : \\
&\quad \quad \text{TokenAdequate}(\bar{\mu}, e)
\end{aligned}$$

2.3 Set comprehension and predicate

Grounded target (a set of **profiles**, $\subseteq \Pi_{A_\alpha}$, not of occurrences): $\mathcal{G}_{\alpha, A_\alpha} = \{p \in \Pi_{A_\alpha} \mid \text{GoalSchema}(\alpha)(p)\}$, and strict soundness $\text{StrictlySoundReasoning}_\chi(e) := \text{ReasoningPathAdequate}_\chi(e) \wedge \text{TokenTruthLinked}_\chi(e)$.

2.4 Underbrace

$$\underbrace{O^*(m; A)}_{\text{ground truth, never asserted from discourse}} \neq \underbrace{\hat{p}_{A, \alpha, t}(m)}_{\text{inferred belief}}$$

2.5 Labeled arrows (world transition vs learner update)

$m_t \xrightarrow{a_t} m_{t+1}$ (occurrence lineage) is not $\theta_t \xrightarrow{U_A(e_t)} \theta_{t+1}$ (learner update).

2.6 Table containing inline math

Object: observation map
Symbol: $\Omega_{A, \alpha, t}$
Non-claim: not the occurrence
Object: inferred profile
Symbol: $\hat{p}_{A, \alpha, t}(m)$
Non-claim: not $O^*(m; A)$
Object: metaorthemma
Symbol: $\bar{\mu}_{e, j}$
Non-claim: not the metaortheme type μ

All notation meanings are those of `docs/notation-registry.yaml`; nothing here introduces a new symbol or claim.